

A Multivariate Analysis of Housing Prices, Metro Access, and Neighborhood Amenities in New Taipei City

By: Beck Peterson

Introduction

This paper looks at what helps explain differences in housing prices in the Sindian District of New Taipei City, Taiwan. More specifically, I study how house price per unit area is related to house age, distance to the nearest metro (MRT) station, and the number of nearby convenience stores. Instead of only looking at one variable at a time, I use a multiple regression model so I can see how each variable is related to price while controlling for the others.

```
df <- df %>%  
  select(  
    age           = `X2 house age`,  
    metro_distance = `X3 distance to the nearest MRT station`,  
    stores        = `X4 number of convenience stores`,  
    price         = `Y house price of unit area`  
  )
```

These variables matter because they each capture a different part of what makes a house more or less valuable. Distance to the MRT is important because easy access to public transportation makes a home more convenient, especially in a city where people rely on transit. The number of convenience stores reflects how many everyday amenities are nearby, which can also make an area more desirable. House age matters too because older houses may lose value over time, but at the same time some older homes may be in more developed or established neighborhoods.

There are good reasons to expect both positive and negative relationships here. I would expect more nearby convenience stores to have a positive relationship with housing prices because people generally value being close to stores and services they use every day. On the other hand, I would expect distance from the MRT station to have a negative relationship with price, since houses that are farther from transit are usually less convenient. House age is a little more complicated. In some cases, older houses may be worth less because of depreciation or maintenance issues, but in other cases older homes may still hold value because of where they are located. Because of that, I allow age to have a nonlinear effect in the model.

The rest of the paper is organized as follows. First, I describe the dataset and the transformations used. Next, I explain the multiple regression model and the basic OLS framework. Then, I discuss the results and interpret the estimated coefficients. Finally, I end with a short conclusion about what the model shows and what its limitations are.

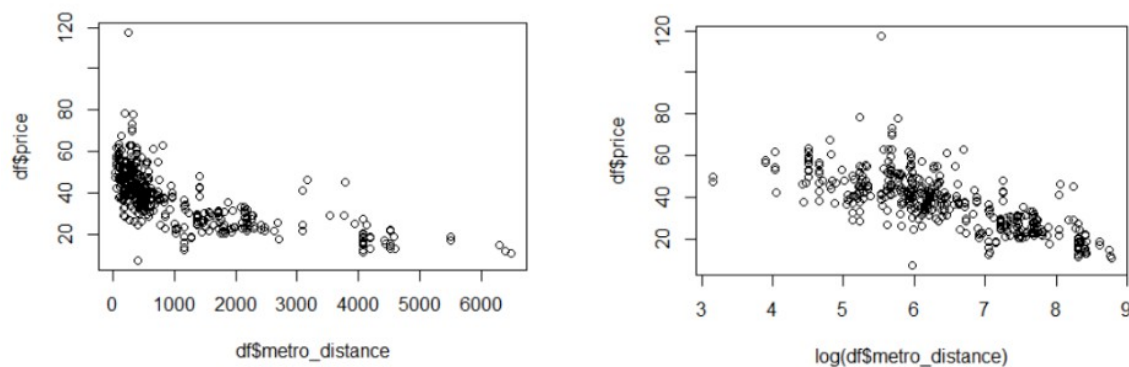
Data

The data come from the Real Estate Valuation dataset, which was also used in the first paper. The full dataset contains 414 observations on housing transactions in New Taipei City, Taiwan. For this paper, I focus on four main variables: house price of unit area, house age, distance to the nearest MRT station, and number of convenience stores.

The observations come from housing transactions that took place in 2012 and 2013. Since the data are cross-sectional, the goal is not to track changes over time but instead to compare houses with different characteristics and see how those differences are associated with price.

I make two main transformations in the model. First, I take the log of metro distance because the raw distance variable is skewed and the relationship with price is not very linear in its original form. Second, I include both age and age squared so the effect of age does not have to be constant across all houses. The dependent variable, house price of unit area, stays in levels in the final model.

Using a log transformation, I obtained a more linear relationship between price and $\log(\text{metro_distance})$ which improved my model.



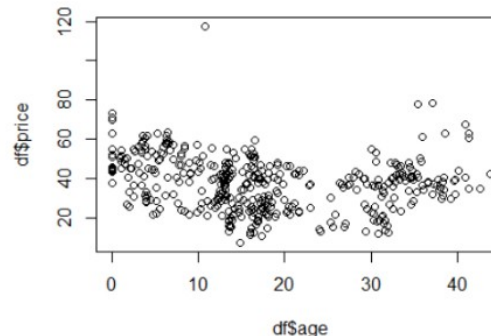
Here, when plotting age vs price, there is a U-shape to the scatterplot. I corrected this by including a polynomial term to age with 2 degrees. Previously, the R-sq was 0.042, and with the polynomial transformation it increases to 0.198.

```
age_mod <- lm(price ~ age, df)
summary(age_mod)$adj.r.squared

## [1] 0.04201891

age_sq_mod <- lm(price ~ age + I(age^2), df)
summary(age_sq_mod)

##
## Call:
## lm(formula = price ~ age + I(age^2), data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -26.542  -9.085  -0.445   8.268  79.961
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  53.450539   1.651624   32.362  <2e-16 ***
## age         -1.928877    0.193759  -9.955  <2e-16 ***
## I(age^2)      0.042181    0.004689   8.995  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.19 on 411 degrees of freedom
## Multiple R-squared:  0.2015, Adjusted R-squared:  0.1977
## F-statistic: 51.87 on 2 and 411 DF,  p-value: < 2.2e-16
```



For stores, I considered using it as a factor with dummy variables since it is a discrete variable and there were not many options, but it did not improve the model's R-sq at all and made the model more complicated. Plus, there does seem to be a slight positive correlation between the number of stores nearby and price.

Methodology

This paper uses a multiple linear regression model estimated by ordinary least squares, or OLS. In a multiple regression, the idea is to explain one dependent variable using several independent variables at the same time. This is helpful because in real life house prices are not determined by just one thing. A model with multiple regressors lets me control for other important factors and gives a cleaner interpretation of each coefficient.

The model used in this paper is:

$$price_i = \beta_0 + \beta_1 age_i + \beta_2 age_i^2 + \beta_3 stores_i + \beta_4 \log(metro_distance_i) + u_i$$

In this equation, the house price per unit area is the dependent variable. The OLS estimator chooses the coefficients that minimize the sum of squared residuals, which means it finds the line that best fits the data by making the prediction errors as small as possible overall.

For the OLS estimators to be unbiased, the standard regression assumptions need to hold. The most important one is the zero conditional mean assumption, which means the error term cannot be related to the explanatory variables once the included regressors are taken into account. In other words, there cannot be an important missing factor in the error term that is correlated with age, stores, or metro distance. If that happens, then the coefficients can be biased.

Results

The estimated regression is:

$$\widehat{price} = 85.5659 - 0.8086(age) + 0.01488(age^2) + 0.7677(stores) - 6.7278\log(metro_distance)$$

With 414 observations, an R^2 of 0.5951, and an adjusted R^2 of 0.5912. This means the model explains about 59.5% of the variation in house price per unit area in the sample, which is a pretty solid fit for cross-sectional housing data.

The coefficient on stores is positive, which matches what I expected. Holding the other variables constant, one additional convenience store nearby is associated with about a 0.77 increase in house price per unit area. This suggests that people value neighborhoods with more nearby amenities and daily convenience.

The coefficient on $\log(\text{metro distance})$ is negative, which also makes sense. Holding age and stores constant, houses that are farther from the MRT station tend to have lower prices. Since distance is in log form, the interpretation is in percentage terms. A 1% increase in metro distance is associated with about a 0.067 decrease in house price per unit area. Overall, this supports the idea that access to public transportation is an important part of housing value in this market.

The age variables show that the relationship between age and house price is not just a straight line. The coefficient on age is negative, but the coefficient on age squared is positive. That means age has a curved effect. For younger and middle-aged properties, getting older tends to lower price, which fits the usual depreciation story. But because the squared term is positive, that negative effect gets smaller as age increases and may eventually turn positive. This could mean that some older houses are located in more established neighborhoods that help support their value. Including both age and age squared makes the model more realistic than using age alone.

The intercept is 85.5659, which is the predicted house price when all explanatory variables are equal to zero, with metro distance equal to one meter so that the log term equals zero. Technically that is the correct interpretation, but in practice the intercept is not the most meaningful part of the model because that combination of values is not very realistic. The slope coefficients are much more useful because they describe how price changes as the explanatory variables change.

```

full_mod <- lm(price ~ age + I(age^2) + stores + log(metro_distance), df)
summary(full_mod)

##
## Call:
## lm(formula = price ~ age + I(age^2) + stores + log(metro_distance),
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -33.669  -4.875  -0.882   3.788  75.380
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    85.565861   3.990034   21.445 < 2e-16 ***
## age           -0.808625   0.150831   -5.361 1.39e-07 ***
## I(age^2)        0.014882   0.003639    4.090 5.20e-05 ***
## stores          0.767713   0.201514    3.810 0.00016 ***
## log(metro_distance) -6.727761  0.558888  -12.038 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.7 on 409 degrees of freedom
## Multiple R-squared:  0.5951, Adjusted R-squared:  0.5912
## F-statistic: 150.3 on 4 and 409 DF,  p-value: < 2.2e-16

```

At the same time, I do not want to overstate what this model can do. There are definitely stories that could bias the estimates. The biggest issue is omitted variable bias. House prices are also affected by things like square footage, building quality, number of bedrooms, renovation status, neighborhood safety, school quality, and exact location. If any of these missing factors are correlated with metro distance, stores, or age, then the estimated coefficients may partly reflect those omitted effects. For example, houses closer to MRT stations may also be in higher-demand neighborhoods in general, so the metro coefficient could be capturing more than transit access alone. In the same way, the number of convenience stores may be standing in for overall neighborhood quality, not just access to stores themselves.

Because of that, the results should be interpreted as associations, not perfect causal effects. The multiple regression model is still a big improvement over the single-variable model because it controls for more relevant factors, but it does not completely eliminate all possible bias.

Conclusion

This paper examined how house price per unit area is related to house age, distance to the nearest MRT station, and the number of nearby convenience stores in the Sindian District of New Taipei City. Using a multiple regression model, I found that houses farther from the MRT station tend to have lower prices, while houses with more nearby convenience stores tend to have higher prices. I also found that the relationship between house age and price is nonlinear, which suggests that age affects housing values in a more complicated way than simple depreciation alone.

Overall, the results suggest that accessibility and neighborhood amenities both play an important role in explaining housing values. Even though the model has limits and may still suffer from omitted variable bias, it gives a more complete picture than a simple one-variable regression. That makes it a useful way to study how multiple housing characteristics are related to price at the same time.